The Data

You'll work with two datasets. One covers how countries around the world produce and use electricity. The other covers how coal is bought, burned, and stored across the United States. Together they show the global shift in power generation alongside the market for the fuel that shift is moving away from.

Dataset 1: U.S. Coal Consumption and Quality (EIA) (Beginner and Advanced)

Each row is one U.S. state (or region, or the national total), one type of user, and one year from 2020 to 2025. The types of user include electric power plants, industrial users, and coke plants. The data shows how much coal each group used, what they paid, and what kind of coal it was.

Volume

- `consumption`: coal burned during the year, in thousands of short tons.
- `receipts`: coal delivered during the year, in thousands of short tons.
- `stocks`: coal sitting in stockpiles at year-end. Rising stockpiles can mean demand is falling faster than purchases are being cut.

Price

- `price`: average price paid per short ton, delivered.

Quality

- `heat-content`: how much energy the coal holds (Btu per pound). Higher-energy coal is generally worth more.
- `sulfur-content`: sulfur as a percentage of weight. High-sulfur coal is more polluting and costlier to burn under environmental rules.
- `ash-content`: ash as a percentage of weight. More ash means more waste to handle.

Each measure comes with a matching units column. Check it before comparing numbers.

Things to Watch For

- **Missing values are blank, not zero.** A blank means the data wasn't available. Don't treat it as "none."
- **Shares are based on production, not use.** A country that imports clean power from a neighbor doesn't get credit for it in its own clean share.
- **Don't double-count.** The coal data includes state rows, regional rows, and a U.S. total, as well as a "total" sector alongside the individual sectors. Pick one level before adding anything up.
- **Recent years may be revised.** Figures for 2025 are often preliminary.

- **Different scopes.** The electricity data is global. The coal data is U.S.-only. Use the coal data to examine the U.S. coal market directly, and the global data for the broader context.

Dataset 2: Global Electricity by Country (Ember) (Advanced)

Each row is one country in one year, from 2020 to 2025. It shows how much electricity the country made, how much it used, and how much came from clean versus fossil sources.

Who and when

- `country`, `iso3`, `continent`: the country, its three-letter code, and its region.
- `year`: the year the data describes. Not every country has data for every year.

How much power

- `generation_twh`: total electricity produced inside the country.
- `demand_twh`: total electricity the country needed. This is what it produced plus whatever it bought from neighbors.
- `net_imports_twh`: electricity bought from other countries minus electricity sold to them. A negative number means the country sells more than it buys.
- `net_import_share_pct`: the share of the country's electricity that came from abroad.
- `demand_per_capita_mwh`: electricity used per person. This is a rough signal of how developed and industrialized a country is.

TWh (terawatt-hours) is the standard unit for national electricity totals. One TWh is roughly enough to power 90,000 U.S. homes for a year.

Where the power comes from

- `clean_generation_twh`: power from renewables plus nuclear.
- `renewable_generation_twh`: power from hydro, wind, solar, bioenergy, and other renewables. This excludes nuclear.
- `wind_solar_generation_twh`: power from wind and solar only, the fastest-growing sources.
- `fossil_generation_twh`: power from coal, gas, and oil.
- `clean_share_pct` and `wind_solar_share_pct`: those sources as a percentage of everything the country generated.

Emissions

- `power_emissions_mtco2e`: estimated greenhouse gases from the power sector, in millions of tonnes. These are modeled estimates, not direct measurements.

- `carbon_intensity_gco2e_kwh`: how dirty each unit of electricity is. Lower is cleaner. This is the easiest way to compare countries of very different sizes.

Year-over-year change

- `demand_change_twh`, `clean_change_twh`, `fossil_change_twh`: how much each quantity rose or fell from the previous year.
- `clean_share_change_pp`: how many percentage points the clean share moved. Going from 30% to 33% is +3 points.
- `clean_growth_surplus_twh`: clean growth minus demand growth. A positive number means new clean power more than covered the country's new demand.
- `clean_growth_coverage_pct`: the share of new demand that new clean power covered. Over 100% means clean growth outpaced demand growth.
- `growth_status`: a plain-language label for the same idea. It says whether demand grew, and if so, whether clean power kept up.
- `has_previous_year`: equals 1 if the country also has data for the prior year, which is needed to calculate any of these changes.